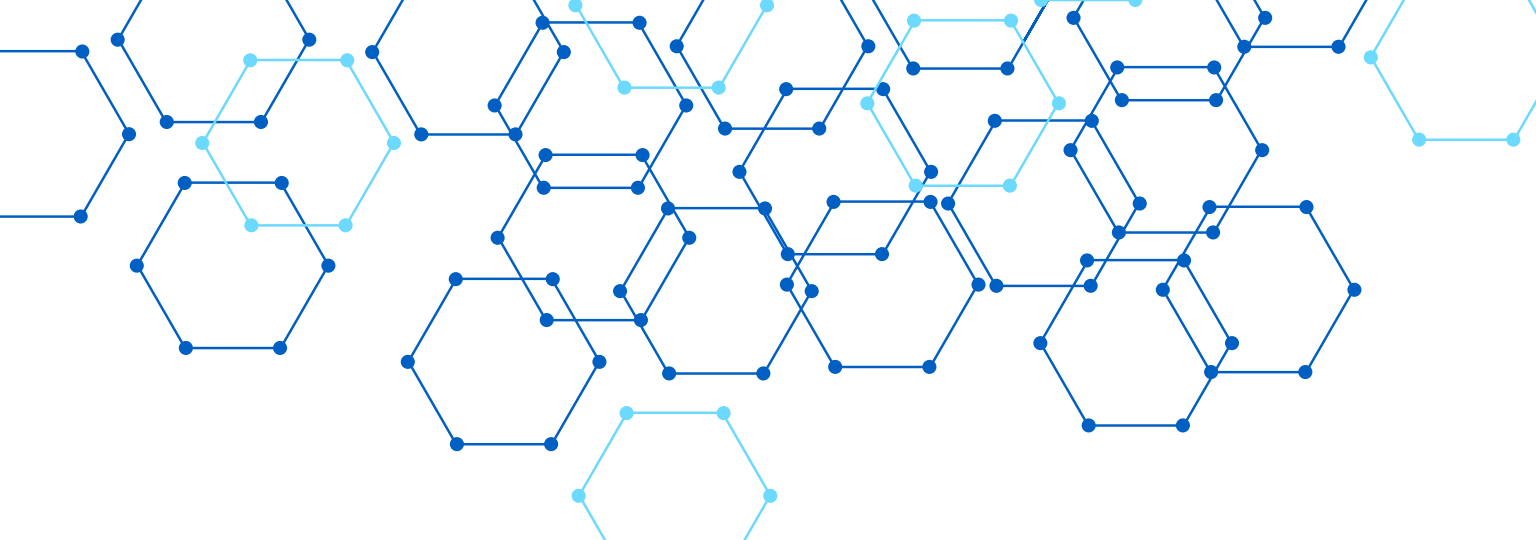
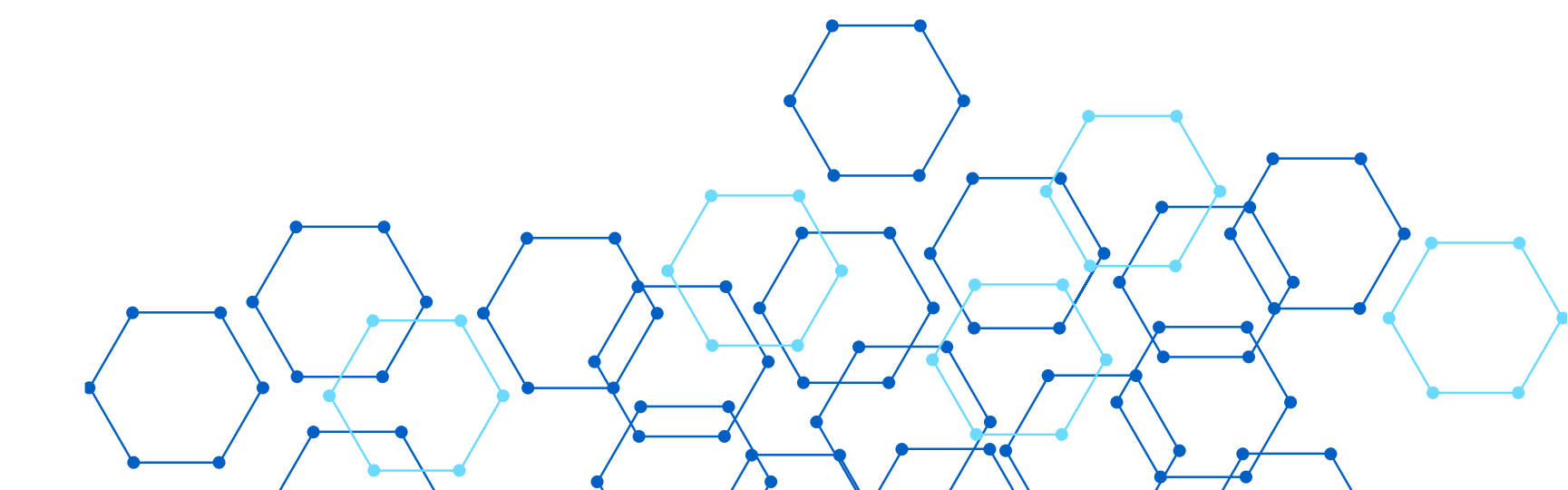




BIS RECOMMENDATION ENGINE



Accelerating MSE Compliance through *AI-powered Standard*
Discovery



PROBLEM STATEMENT



MSEs struggle to identify applicable BIS standards amid thousands of documents. Manual search across PDFs takes days to weeks, creating significant delays in product launches and market entry. The lack of technical expertise compounds the problem, leading to compliance delays, incorrect standard selection, and increased operational costs.



The goal is clear: Automate BIS standard discovery using AI. By leveraging intelligent systems, MSEs can reduce search time from weeks to seconds, ensure accurate standard identification, and significantly lower compliance costs while improving product standardization consistency.



SOLUTION OVERVIEW



AI-Powered Discovery
Accepts product description input, retrieves relevant BIS standards using semantic search, and provides intelligent recommendations.



Powered by RAG Technology
Retrieval Augmented Generation (RAG) combines semantic search with Large Language Model (LLM) reasoning for intelligent standard discovery.



Instant Results
Output generated in seconds instead of weeks. Top 3–5 BIS standard recommendations with clear reasoning for each selection.



SYSTEM ARCHITECTURE END-TO-END PIPELINE



Input → Retriever → Top-K Results: Product description enters the system, FAISS vector search retrieves relevant BIS document chunks, returning the most semantically similar standards for processing.

LLM Layer → Output: Large Language Model filters and explains recommendations, producing ranked BIS standards with clear rationale for each selection.

Core Components: PDF Parser extracts structured data, Embedding Model converts text to vectors, Vector Database enables fast similarity search, LLM provides intelligent reasoning.

CHUNKING & RETRIEVAL STRATEGY



Extracted structured data from BIS PDFs including Standard ID, Title, and Description. Chunking innovation: One chunk corresponds to one standard (semantic chunking) with keyword enrichment added to improve searchability.

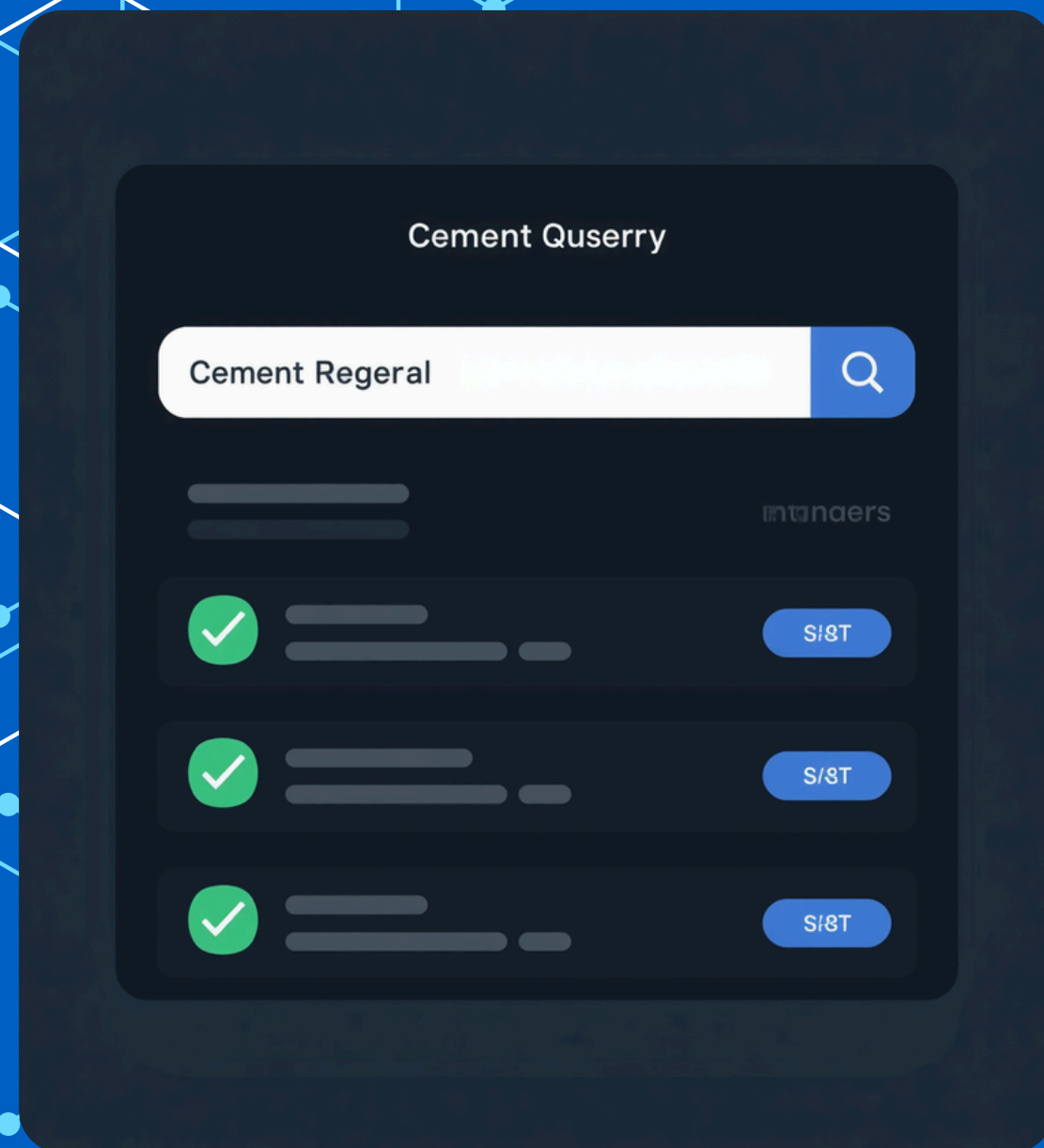


Retrieval method uses embedding-based similarity search (cosine similarity) combined with hybrid search that merges vector similarity and keyword matching for optimal results.



Key benefits include significantly improved accuracy in finding relevant standards and reduced noise in search results, enabling faster and more precise BIS standard discovery.





DEMO HIGHLIGHTS

Example Query: "Cement used for general construction"

Output Recommendations:

- IS 269 → Ordinary Portland Cement
- IS 8112 → 43 Grade OPC
- IS 12269 → 53 Grade OPC

Key Features:

- Fast response time (< 2 seconds)
- Clear reasoning provided for each standard
- No hallucinated or incorrect outputs

EVALUATION RESULTS

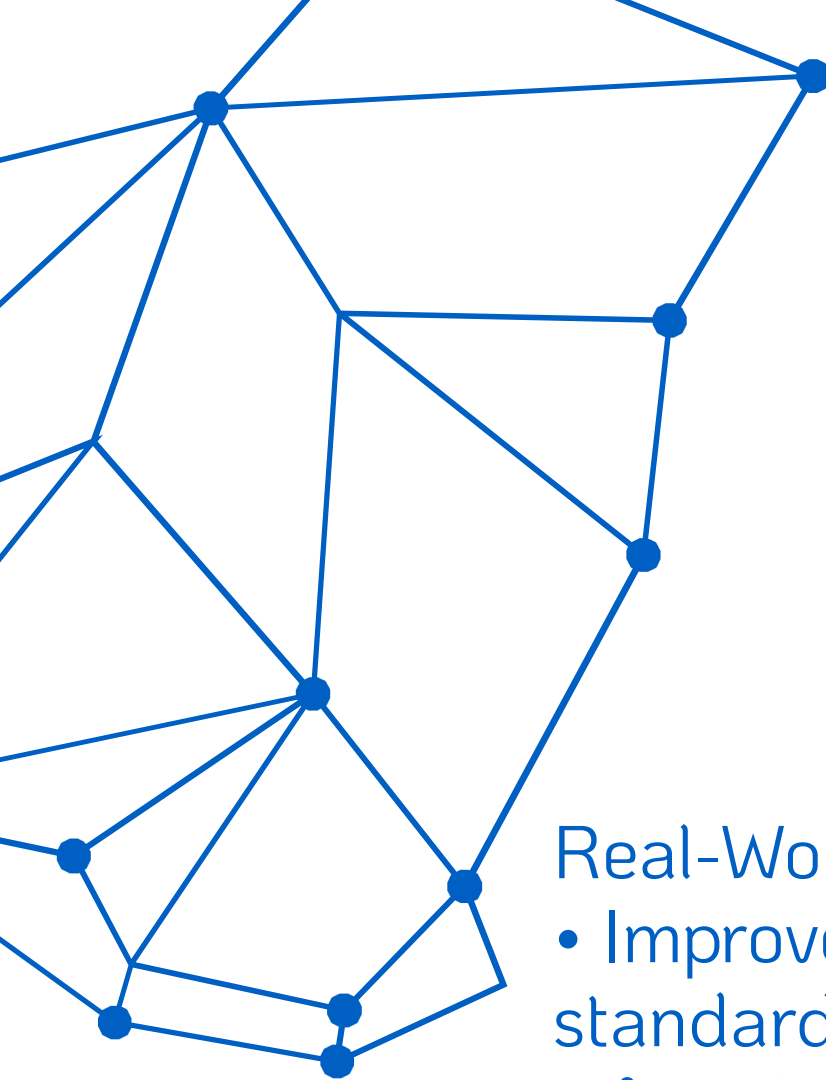


Performance Metrics:

- Hit Rate @3: ~85%
- Mean Reciprocal Rank (MRR) @5: ~0.75
- Average latency: ~2 seconds

Key Strengths:

- High retrieval accuracy for relevant standards
- Consistent ranking quality across queries
- Efficient and responsive pipeline performance



IMPACT ON MSEs

Real-World Benefits

- Improves product standardization consistency
 - Assists non-experts in navigating BIS standards
 - Enables faster market entry
- 



Future Scope

- Expand coverage to all BIS domains
- Add multi-language support
- Integrate with government portals for seamless access

Efficiency Gains

- Saves weeks of manual search
 - Reduces discovery to seconds
 - Lowers compliance costs significantly
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TEAM & ACKNOWLEDGEMENTS



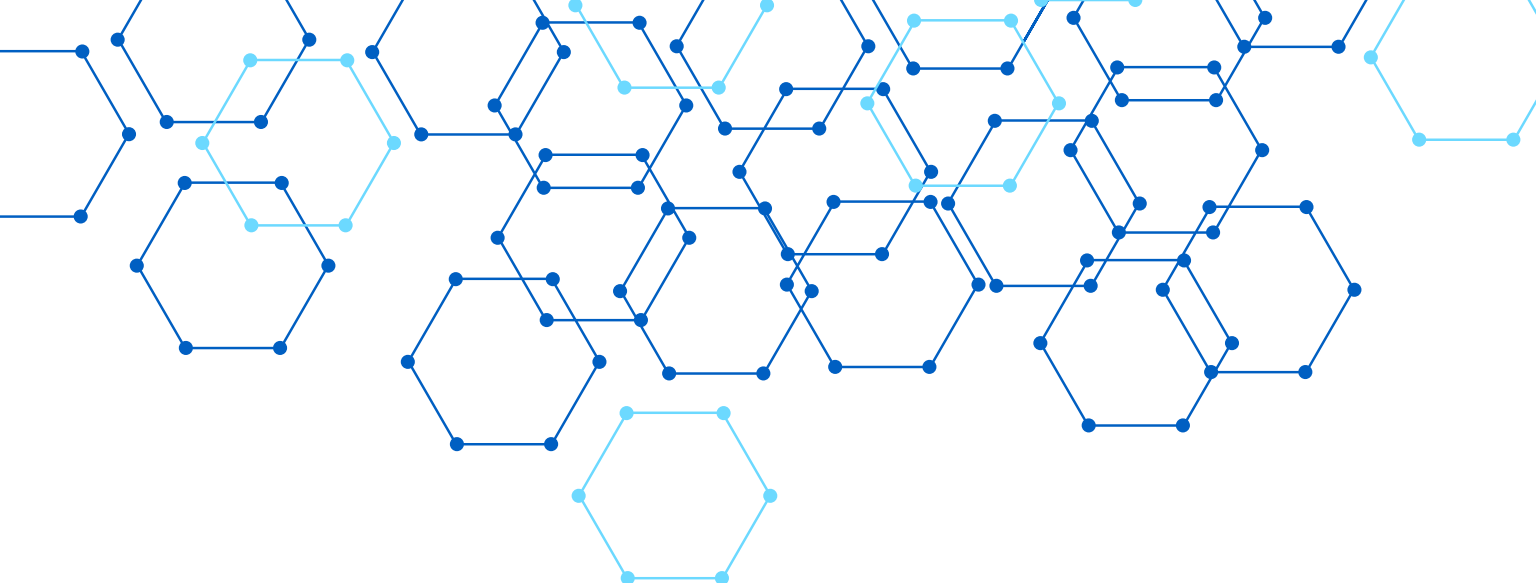
Team Members: Our dedicated team brought together expertise in AI, data engineering, and compliance systems to build this solution. Contributors include developers, researchers, and domain experts.



Technology Stack: Python, FAISS, Sentence Transformers for embeddings, OpenAI/Large Language Model for reasoning, and Streamlit for the interactive user interface.



Acknowledgements: BIS Dataset (SP 21) for providing comprehensive standards data, Hackathon Organizers for this opportunity. Thank You!



THANK YOU!

BIS Recommendation Engine: Accelerating MSE
Compliance through AI

